

SANGRAM KUMAR NAHAK

SENIOR GIS ENGINEER | LiDAR & GEOSPATIAL

CONTACT



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CORE SKILLS

*LiDAR: Airborne & UAV LiDAR, point-cloud processing, classification, QA/QC
*Classification: Ground, building, vegetation, water, noise, HAG, feature extraction
*Surface: DTM/DSM, TIN, breaklines, surface & volume modelling, contours
*Powerline: Corridor mapping, centerline, catenary analysis, danger-zone workflows
*Automation: TerraScan macros, classification macros, Python workflow automation
*GIS: ArcGIS, QGIS, Global Mapper, MicroStation, AutoCAD

SOFTWARE

TerraSolid / TerraScan
MicroStation
ArcGIS
QGIS
Global Mapper
AutoCAD
Agisoft Metashape
HTML / CSS / JavaScript
Python

EDUCATION

Master of Computer Applications
Amity University | Jun 2023 – Mar 2025
Bachelor of Arts – English Language & Literature
Berhampur University | Aug 2016 – May 2019

PROFILE

GIS and LiDAR professional with 5+ years of experience in airborne and UAV LiDAR processing, geospatial QA/QC, surface modelling, powerline corridor mapping, photogrammetry and WebGIS. Experienced in point-cloud classification, TerraScan macro automation, DTM/DSM/TIN production, catenary analysis and client-facing technical delivery, including international project exposure in Qatar and UK infrastructure work.

WORK EXPERIENCE

GEOKNO INDIA PRIVATE LIMITED

Geographic Information Systems Consultant

Nov 2025 – Present | Hyderabad

- Project delivery and technical coordination for international stakeholders; tracked team efficiency and schedules and supported on-time delivery.
- Executed Survey of India (SOI) GIS workflows including georeferencing, attribute editing, land partitioning and map production.
- Processed LiDAR point clouds including ground, building, feature and water classification; created breaklines, TINs, volume surfaces and DTM/DSM outputs.
- Performed powerline corridor workflows including metadata extraction, centerline creation, catenary analysis and automated danger-zone identification.
- Converted corridor outputs to CAD/Global Mapper formats and produced final cartographic deliverables in ArcGIS and QGIS.

ATOMGEO PRIVATE LIMITED

Geographic Information Systems Engineer

Sep 2025 – Nov 2025 | India

- Performed GIS/LiDAR quality assurance and analysis as a final technical quality gate for complex spatial datasets.
- Validated LAS/LAZ metadata, point density, coordinate systems and class statistics.
- Conducted outlier detection, noise removal and spatial discrepancy analysis using MicroStation and Microsoft Excel before client handover.

NAKS HATECH PVT LTD

Geographic Information Systems Engineer

Aug 2023 – Sep 2025 | India

- Led engineering workflows and mentored engineers across photogrammetry, orthorectification and LiDAR production activities.
- Processed drone point clouds, optimized GCP networks and generated colorized LAZ deliverables through orthophoto alignment in Agisoft Metashape.
- Developed classification macros for ground, vegetation and building classes and performed 2D/3D feature extraction for UK infrastructure projects.
- Supported powerline processing through flight-line alignment and classification QA/QC; worked with TerraMatch for flight-line matching and trajectory adjustment.

KHATIB & ALAMI

GIS Engineer | Senior Technician – International Deployment

Apr 2024 – Dec 2024 | Qatar

- Led airborne LiDAR processing for large-scale urban mapping projects in Qatar, maintaining project-defined geometric and vertical accuracy requirements.
- Performed advanced strip alignment, point-cloud registration and surface modelling to produce clean, client-ready DTM/DSM outputs.
- Worked directly with public-sector stakeholders, including Qatar's Ministry of Municipality and Environment (MME).

SPECTRA ENGINEERING SERVICES LTD.

LiDAR | Team Leader

Mar 2022 – Aug 2023 | Bengaluru

- Fast-tracked to Team Leader; managed project timelines, production workflows and QA/QC for external vector and LiDAR deliverables.
- Led environmental feature extraction workflows including standing/flowing water breakline generation and forestry analysis involving tree roots, trunks and branches.
- Supported WebGIS spatial-data hosting and 3D visualization workflows using HTML, CSS, JavaScript and Python automation.

GEODESY MAPPING SOLUTIONS PRIVATE LIMITED

LiDAR Engineer

Sep 2021 – Mar 2022 | India

- Performed QA/QC of airborne and drone LiDAR datasets for UK infrastructure projects, supporting DTM/DSM production.
- Executed statistical/manual filtering, flight-line matching and noise removal to meet project precision requirements.
- Provided onsite technical support and demonstrated LiDAR integration with ArcGIS, Global Mapper and Google Earth.

NAKS HATECH PVT LTD

LiDAR Engineer

Oct 2020 – Sep 2021 | India

- Processed airborne and drone LiDAR datasets, including trajectory correction and flight-line matching for large-area and corridor mapping projects.
- Developed foundational macros for ground-seed generation, height-above-ground logic and low/medium/high vegetation classification.
- Supported point-cloud preparation through noise filtration, feature extraction and DTM/DSM production using MicroStation and LiDAR software

TECHNICAL FOCUS

GIS and LiDAR professional with 5+ years of experience in airborne and UAV LiDAR processing, geospatial QA/QC, surface modelling, powerline corridor mapping, photogrammetry and WebGIS. Experienced in point-cloud classification, TerraScan macro automation, DTM/DSM/TIN production, catenary analysis and client-facing technical delivery, including international project exposure in Qatar and UK infrastructure work.